

Andhra Pradesh Rolls Out an Electronic Public Distribution System Using Open Source



Totally based on biometric authentication, the electronic public distribution system project initiated and implemented by the state government of Andhra Pradesh has been developed on PostgreSQL, the open source database.

It was in 2014 that the Andhra Pradesh government decided to go digital and opted for an electronic solution for its PDS (public distribution system). While the initial model was tested in a few small regions, the final release can handle as many as four million transactions per day during peak time across all the 13 districts in the state—all this powered by PostgreSQL, the open source database.

The Andhra Pradesh (AP) government gave the task of transforming its public distribution system to a team at the National Informatics Centre (NIC). Led by civil supplies commissioner and state informatics officer (SIO) of Andhra Pradesh, K. Rajasekhara, the NIC team preferred US-based

Facts about AP's iconic ePDS project

- Delivers benefits to around 30,000 concurrent users in the state
- Handles over four million transactions per day during peak time
- 100 per cent Aadhaar-enabled system with biometric authentication
- More than 28,000 ePOS devices and many electronic weighing machines are in use

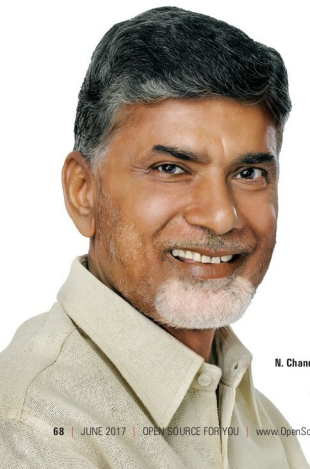
EnterpriseDB to build a progressive ePDS (electronic public distribution system) model, in order to improve the transparency, efficiency and effectiveness of the system. The arrival of the ePDS in the region has also set the stage for next-generation solutions, including eSCM (electronic supply chain management) and ePOS (electronic point of sale). All these developments have led to hyper-scalability, high-availability, reliability and high-concurrency to the entire process of delivering civil supplies in the state, along with low latency.

ePDS: a mission critical application

The Postgres database, which currently holds nearly 38 million records, was deployed right at the beginning of the project. This open source database helped to enable Aadhaar-based biometric authentication in real-time, and offer cashless as well as cardless transactions.

"The application that we have developed with the support of NIC and EnterpriseDB for distributing rations in the region is very mission critical and citizen-centric," Rajasekhara told *Open Source For U*.

The latest Postgres release (version 9.6), which has been powering the distribution system, can be scaled to more than 1.5 million transactions per second (TPS). However, the team is working on parallel data processing features to achieve a higher TPS.



N. Chandrababu Naidu,
chief minister,
Andhra Pradesh